Track 1: A Biochemical Rapid/Spot Testing Kit for the Detection of Synthetic Drugs

Problem Description

Develop a portable, rapid, and cost-effective biochemical testing kit for on-site detection of synthetic and emerging psychoactive substances. The solution should enable quick preliminary screening with high accuracy to support timely forensic examination and law enforcement operations.

Explanation

The rapid proliferation of novel synthetic drugs presents a critical challenge for law enforcement and border security. Traditional chemical spot tests rely heavily on subjective visual interpretation by officers in the field, which can be easily compromised by poor lighting, human bias, or visual fatigue. Furthermore, these manual tests lack automated digital logging, complicating the forensic chain of custody. A digitized, sensor-based testing device eliminates these issues entirely. By utilizing optical sensors and edge computing to quantify chemical color reactions with mathematical precision, the solution provides police officers with immediate, highly accurate screening results. This allows law enforcement to easily and confidently identify substances on the spot while automatically securing the data for legal proceedings.

Objective

To modernize traditional chemical spot testing, the proposed solution must achieve the following core objectives:

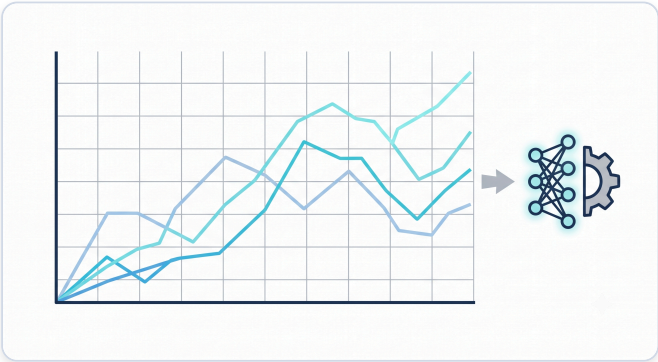
- **Eliminate Visual Bias:** Replace subjective human interpretation of color reactions with precise optical sensors and automated algorithmic analysis.
- **Rapid Field Detection:** Deliver clear and actionable preliminary screening results to officers within a strict 2 to 5-minute timeframe.

- **Secure Digital Logging:** Establish a robust digital chain of custody by automatically logging test results, confidence scores, and cryptographic hashes.
- **Maximized Portability:** Ensure the device maintains a minimal, lightweight hardware footprint suitable for dynamic, on-site law enforcement operations.
- **Scalable Affordability:** Utilize accessible, low-cost edge computing components to ensure the hardware solution is highly reproducible for widespread police deployment.

Expected Features for Evaluation

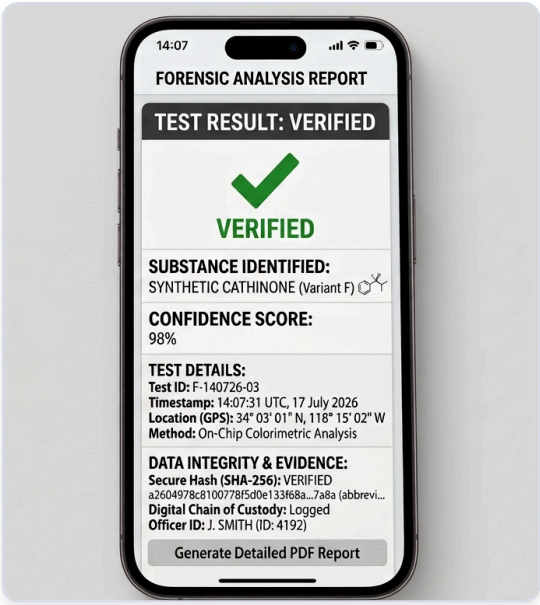
Participants will be evaluated on the successful integration and demonstration of the following features:

- **Microcontroller & Sensor Integration:** Effective integration of an edge processing board (e.g., Arduino or similar) with an optical or color sensor to capture the color properties of the chemical reaction directly.
- **AI/ML Colorimetric Analysis & Extensibility:** Implementation of Machine Learning models to analyze the sensor's raw data stream over time. The AI must be capable of **multi-analyte classification** (differentiating distinct classes like cathinones vs. opioids) from a single scan. The software architecture should also support **model updates** to recognize newly emerging drug analogues without requiring new hardware.



- **Software & Logging Application:** Development of a dedicated companion application or digital dashboard (mobile or desktop) that interfaces with the hardware to display the predicted substance and confidence score. The app

must also **automatically capture GPS geotags** of the seizure location and securely log all operational data.



- **Data Integrity & Cybersecurity:** Application of cryptographic hashing (e.g., SHA-256) and secure data handling to create an immutable digital chain of custody, ensuring that test results, timestamps, GPS coordinates, and sensor logs cannot be tampered with before reaching the forensic database.



- **Speed of Detection:** The complete automated sensor acquisition and classification process must deliver clear results in under 5 minutes.
- **Hardware Optimization & Operator Safety:** The prototype device must feature a streamlined, lightweight design, ensuring it is highly portable. The design must also prioritize **operator safety** by incorporating a sealed or isolated sensor interface that requires minimal sample quantities, protecting officers from toxic exposure.

- **Cost-Effectiveness:** The prototype must prioritize low-cost electronic components, making the hardware solution affordable and highly reproducible for deployment.

Tech Stack:

We do not mandate a specific technology stack. Instead, teams have the freedom to select their preferred tools and frameworks. You will be expected to defend your technical choices during the evaluation, demonstrating how they optimize the solution for performance, security, long-term maintainability, and scalability.

References:

1) Philp, M. & Fu, S. (2018). "A review of chemical 'spot' tests: A presumptive illicit drug identification technique." Drug Testing and Analysis, 10(1), 95-108. Link

2) UNODC - Early Warning Advisory (EWA) on New Psychoactive Substances (NPS). Link

3) SWGDRUG - Scientific Working Group for the Analysis of Seized Drugs, Approved Recommendations (minimum forensic standards for seized-drug identification, including presumptive/colour testing). Link

> ⚠️ **Safety Note for Demonstration:** Under no circumstances are real illegal substances or hazardous reagents permitted at the venue. Teams must use safe, harmless household items (e.g., pH strips, food coloring, baking soda) to simulate their prototype's functionality during the hackathon pitch.